Journal Pre-proof

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PII: S1385-8947(26)07159-7

DOI: https://doi.org/10.1016/j.cej.2026.179698

Reference: CEJ 179698

To appear in:

Received date: 23 December 2025

Revised date: 2 April 2026

Accepted date: 18 July 2026

Atomic-scale Ir decoration on Pt(100) via self-terminated electrodeposition for electrochemical ammonia oxidation

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Abstract

The ammonia oxidation reaction (AOR) suffers from sluggish kinetics and poisoning by nitrogen intermediates, limiting its application in energy systems. Here, we present a self-terminated electrodeposition (SED) strategy to decorate Pt(100) nanosheets with ultra-trace Ir, enabling atomic-scale control of interfacial structure. Pulsed deposition directs Ir preferentially to step sites at low coverage, forming site-isolated Ir domains that maximize activity while suppressing poisoning. Under alkaline conditions, Ir decoration induced a volcano-type dependence of activity on Ir content. The Ir5/Pt NT electrode (5.67% surface Ir) achieved a specific activity of 0.772 mA/cm^2 at 0.60 V_{RHE}, while its integrated AOR charge density was nearly double that of pristine Pt NT, underscoring cumulative enhancement over the full potential window. X-ray photoelectron spectroscopy and electrochemical characterization confirm that this enhancement originates from isolated Ir at step sites, whereas extended Ir domains at higher coverages disrupt this configuration and diminish performance. Density functional theory calculations identify $*NH_2$ coupling as the rate-determining step, with its free energy barrier decreasing from 1.45 eV on Pt to 1.30 eV at Ir–Pt step interfaces due to cooperative stabilization of intermediates. In addition, performance was evaluated in a membrane electrode assembly-type ammonia electrolyzer, where the Ir-decorated anode exhibited improved performance at low cell voltages. Collectively, these findings highlight the mechanistic and structural basis for optimizing AOR catalysts and provide a design principle for high-performance electrocatalysts in ammonia-based energy conversion.

Keywords: ammonia oxidation reaction; Ir-decorated Pt (100) catalysts; Ir–Pt interface; $*NH_2$-$*NH_2$ coupling; self-terminated electrodeposition

1. Introduction

Ammonia oxidation reactions (AOR) are of growing interest due to their relevance in diverse energy and environmental applications, including ammonia sensing [1], wastewater remediation [2,3], direct ammonia fuel cells [4,5], and hydrogen purification via residual ammonia removal for proton exchange membrane fuel cells [6]. Ammonia also serves as a hydrogen carrier, benefiting from its high volumetric energy density [7,8], ease of storage and transportation [9,10], and industrial maturity [11]. Yet, AOR implementation is constrained by sluggish kinetics and catalyst deactivation from strongly adsorbed intermediates [12-14].

Among monometallic catalysts for the AOR, Pt demonstrates the superior NH_x dehydrogenation capacity, which is attributed to its optimal *N adsorption strength [15,16]. Facet-controlled Pt {100} catalysts, such as cubic [17,18] and cuboid [19] morphologies, outperform Pt (111) counterparts due to enhanced *N dimerization on Pt (100) surfaces [20–22]. Pt-based alloy catalysts have been studied to further enhance AOR activity by varying their compositional effects [15]. Elements with stronger (Ru, Rh, Ir, and Pd) or weaker (Au, Ag, and Cu) *N adsorption than Pt have been investigated in this context [23]. Ir, in particular, has been widely alloyed with Pt to improve the AOR activity [24–28]. Ir improves the onset potential owing to its stronger interaction with *N than with Pt, but excessive Ir induces N_{ads} poisoning and degrades peak activity [24–28].

Atomic-scale design strategies for AOR catalysts remain scarce beyond bulk alloy systems [29–31]. Liu et al. investigated Pt monolayers on various noble metal supports, including Ru, Rh, Pd, Ir, and Au, and found that Pt supported on Au exhibited the highest specific AOR activity [29]. The activity of Pt overlayers on Au was highly dependent on Pt coverage [30]. A submonolayer coverage of 12–36% achieved more than twice the mass activity compared to a full monolayer

[30]. Further studies by the same group revealed that variations in Au core morphology also affected the AOR performance of the Pt overlayer [31]. Nevertheless, the high cost of Au limits the practical viability of the Pt-on-Au configuration.

In contrast, atomic-scale Ir decoration on Pt electrodes has been reported to enhance AOR activity [32,33]. Fomena et al. demonstrated that a nanostructured Pt (100) film modified with an Ir sub-monolayer enhanced the AOR onset potential and staility when Ir coverage was below 13% [32]. Siddharth et al. reported that trace amounts of Ir decoration (< 2%) increased the AOR activity of Pt nanocubes by more than two-fold [33]. Theoretical calculations have revealed that decorated Ir atoms facilitate stronger $*NH_3$ adsorption and lower the energy barrier for the *NH formation step [33]. These results highlight the promise of atomic-scale Ir decoration for enhancing AOR activity. However, limited electrode performance remains a major barrier to practical application. A detailed understanding of the AOR mechanism, particularly under conditions requiring precise compositional control, remain elusive. Further investigation is needed to establish precise structure–activity relationships.

Among various approaches to depositing Pt or Ir at atomic scale, self-terminated electrodeposition (SED) enables the wet atomic layer deposition of Pt and Ir under ambient conditions through a single-step process [34–37]. In this approach, hydrogen adsorbed during the deposition serves as a self-limiting agent, preventing further reduction and coordination of Pt or Ir ions on the surface [34,35]. Our previous studies demonstrated that metallic Ir deposition via Ir SED is feasible on both flat Au substrates and porous Au transport layers [35,36]. However, this method is limited by the need for costly and catalytically inactive Au, which restricts its practical viability.

In this study, we present a SED strategy to achieve nanogram-scale Ir decoration on Pt(100) electrodes with precise compositional control through pulsed operation. Ir deposition initiates at step sites and extends onto terraces, providing a continuous range of surface coverages. This platform enables us to directly correlate interfacial structure with ammonia oxidation performance under alkaline conditions. We further demonstrate that isolated Ir decoration at step sites maximizes specific activity, while higher coverages lead to performance loss. To further elucidate the mechanistic role of interfacial Ir step, we also employ density functional theory (DFT) calculations to analyze the electronic and energetic implications of Ir–Pt configurations in NH_x activation steps. These insights establish design principles for efficient ammonia oxidation catalysts and provide guidance for advancing ammonia-based energy conversion technologies.

2. Experimental

2.1. Preparation of Pt substrate

The Pt substrate was electrochemically deposited on carbon paper (CP) at room temperature under normal atmospheric pressure. The deposition was performed using a standard three-electrode cell setup connected to a potentiostat (Autolab PGSTAT302N, Metrohm). Bare CP (MGL 280, Avcarb) encased in a custom Teflon cell with an area of 1.44 cm^2 to the deposition solution acted as the working electrode. A Pt wire and a saturated calomel (SCE, KCl-saturated) were employed as the counter and reference electrodes, respectively. The electrodeposition solution consisted of K_2PtCl_4 (99.9%, Alfa Aesar) as the Pt precursor and either 1.27 M HCl (ACS reagent grade, Sigma Aldrich) or 0.5 M H_2SO_4 (95.0%, JUNSEI) as the supporting electrolyte. The electrodeposition conditions, including Pt precursor concentration, potential, and duration, were varied. Prior to Pt deposition, dissolved oxygen was removed from the electrolyte by bubbling with Ar or N_2 gas for 30 min.

2.2. Ir SED on Pt substrate

The deposition bath for Ir SED comprised 0.5 mM K_3IrCl_6 (99.9%, Sigma-Aldrich) and 500 mM Na_2SO_4 (99.0%, Alfa Aesar). The solution pH was set to 4.0 using diluted H_2SO_4, and the Ir SED process was performed at 50 °C in a double-jacketed glass cell connected to a thermostat. The Pt substrate submerged in the N_2-purged deposition solution (exposed area = 0.5–1.0 cm^2) served as the working electrode. A Pt plate was used as the counter electrode, and a mercury sulfate electrode (MSE, K_2SO_4-saturated) with a fritted junction containing 500 mM Na_2SO_4 (pH 4.0) was used as the reference electrode. The deposition pulses were applied repeatedly using a potentiostat.

2.3. Characterization

The sample morphologies were examined using field-emission scanning electron microscopy (FESEM, Carl Zeiss, SIGMA) and transmission electron microscopy (TEM, JEM-2100F, JEOL Ltd.). TEM samples were prepared by dispersing the catalyst in isopropyl alcohol via sonication, followed by drop-casting onto Cu grids. Surface compositions were characterized using X-ray photoelectron spectroscopy (XPS, K-alpha+, Thermo Fisher Scientific). The crystal structures were assessed using selected-area electron diffraction (SAED), high-resolution TEM (HRTEM), and X-ray diffraction (XRD, New D8-Advance, Bruker). The metal content was determined using a weighing technique supplemented with inductively coupled plasma mass spectroscopy (ICP-MS, NexION 300, PerkinElmer).

Hydrogen underpotential deposition (H_{UPD}) measurements were conducted using cyclic voltammetry (CV) in a glass cell connected to a potentiostat. The prepared electrodes served as the working electrodes and a Pt coil was used as the counter electrode. SCE or Hg/HgO (1.0 M NaOH) were used as reference electrodes in Ar-purged 0.5 M H_2SO_4 or 1.0 M KOH, respectively. The H_{UPD} charge density was determined by averaging the sum of the hydrogen adsorption/desorption charge densities from CV. The roughness factor (R_f) of the electrode was calculated based on the H_{UPD} charge density of the flat $metal_{Pt+Ir}$ monolayer (210 μC/cm^2).

2.3.1. AOR activity test

Electrochemical performance was evaluated using a glass cell connected to a potentiostat. The prepared electrodes were used as working electrodes, with a Pt coil serving as the counter electrode and Hg/HgO (1.0 M NaOH) as the reference electrode. The activity for AOR was assessed through CV in an Ar-purged electrolyte of 0.1 M NH_3 (99.9%, Sigma-Aldrich) and 1.0 M KOH (93.0%, DAEJUNG). For the conversion of potential to the reversible hydrogen electrode (RHE), the test was carried out in H_2-saturated 0.1 M NH_3 and 1.0 M KOH using a Pt wire and coil as the working and counter electrodes, respectively. The CV measurements were performed at a scan rate of 1.0 mV/s. The thermodynamic potential for hydrogen evolution was determined by averaging the potentials recorded at zero current during both negative and positive scans. The mass and specific activities were calculated by dividing the geometric AOR activity by the total metal loading and R_f obtained under the condition of 1.0 M KOH. Reaction products were not directly analyzed in this study. Multiple nitrogen-containing species (e.g., N_2, NO, N_2O, and NO_2) can form depending on the applied potential. This study primarily focuses on the low-potential region ($\leq$ 0.6 V_{RHE}), where N_2 is widely reported as the dominant product for ammonia oxidation on Pt-based catalysts [12].

2.3.2. Ammonia Electrolyzer Measurements

The membrane electrode assembly (MEA) was fabricated using prepared electrode as the anode and a 40 wt% Pt/C catalyst (0.5 mg cm^{-2}) supported on carbon paper as the cathode. An anion exchange membrane (Sustainion®, X37-50 Grade RT) was employed in the MEA configuration. Prior to use, the membrane was immersed in 1.0 M KOH for at least 24 h at room temperature. The geometric area of both electrodes was 1 cm^2.

The MEA cell was operated at 80 °C. The catholyte consisted of 1.0 M KOH and was supplied at a flow rate of 5 mL min^{-1}. The anolyte contained 1.0 M KOH + 1.0 M NH_4OH and was also delivered at 5 mL min^{-1}. The anolyte was prepared immediately prior to use to minimize NH_3 evaporation.

2.3.3. Computational Details

DFT calculations were performed using the Vienna ab-initio simulation package (VASP) to investigate the AOR mechanism on Ir steps decorated on Pt surfaces [38]. The electron exchange-correlation functional is described using the generalized gradient approximation (GGA) of revised Perdew-Burke-Ernzerhof (rPBE) functional [39,40]. The core and the valence electrons are described by the projector augmented wave (PAW) potentials and the plane-wave basis set with a cutoff energy of 520 eV [41,42]. The Gamma-centered K-points sampling method was used with the mesh of 4 x 3 x 1. All slab models are relaxed to the electronic convergence criterion of 10^{-5} eV and the atomic force tolerances of 0.03 eV Å^{-1}.

To model the Ir steps decorated on Pt surface, the (100) surface of Pt face centered cubic (FCC) structure was built up with three layers, measuring 11.25 and 16.87 Å in the x-and y-direction, respectively, with a vacuum region of 26 Å in the z-direction. The lateral supercell dimensions were selected to be sufficiently large to avoid artificial interactions between periodic images of the adsorbates under periodic boundary conditions. [43,44] Each layer consisted of 24 Pt atoms. In addition, Ir step layer was constructed by covering half of the surface, which consists of 12 Ir atoms from the top layer. In addition, Pt step model was constructed in the same manner, where the top layer was partially covered with 12 Pt atoms occupying half of the top surface area, analogous to the Ir step layer. For all calculations, the bottom layer of Pt was fixed to their bulk position. Bader charge analysis was performed to evaluate the charge of Ir step supported on Pt surface [43].

The free energies of all species (N_xH_y) were referenced to those of gaseous NH_3 and H_2. Gibbs free energy diagram was implemented by modelling proton-electron transfers steps using the computational hydrogen electrode approach. Zero-point energy (ZPE) was estimated based on the harmonic oscillator approximation, using the vibrational frequencies via second-order finite difference calculations of atomic forces with a step-size of 0.02 Å. Entropy contributions of the adsorbed species were evaluated by considering translational, rotational, and vibrational degrees of freedom [44].

3. Results and discussion

3.1. Development of Ir SED technique on Pt surface

Pt electrodeposition was conducted in 2 mM K_2PtCl_4 and 1.27 M HCl electrolyte using a CP as the substrate (**Figure 1a,b**). The morphological features of the prepared Pt on CP were identified by FESEM (**Figure 1c-e**). At low magnification (**Figure 1c**), the Pt deposit appeared to uniformly cover the carbon fibers in a film-like manner, whereas its nanosheet-like texture was confirmed in the high-magnification images (**Figure 1d,e**). Based on this morphology, the Pt electrode was designated as Pt-NT. XRD analysis (**Figure 1f**) confirmed its bulk polycrystalline nature.

As shown in **Figures 2a and S1**, the feasibility of Ir SED on Pt NT was investigated by CV in a 500 mM Na_2SO_4 electrolyte (pH 4.0) with varying concentrations of K_3IrCl_6. In previous studies for Ir SED on Au substrate, Ir deposition was thermally activated above 40 ℃ [35,36]. To accelerate the sluggish reduction kinetics of Ir^{3+}, Ir SED was performed at 70 ℃, yielding Ir coverages from 8.1% to 73.5% on porous Au substrates [36]. In this study, the Ir deposition was conducted at 50 °C to enable lower coverage with minimal activation of Ir^{3+}reduction.

At 3.0 mM K_3IrCl_6 electrolyte, the CV revealed features characteristic of a self-terminating phenomenon under limited Ir^{3+} flux. The reduction of Ir^{3+} on the Pt NT surface began at an onset potential of approximately –0.35 $V_{MSE,}$ and the reduction current density gradually increased up to –0.60 V_{MSE}. Beyond this point, the reduction current density remained nearly constant at approximately –0.4 mA/cm^2, even as the potential continued to extend negatively. At potentials beyond –1.00 V_{MSE}, the Ir^{3+} reduction was quenched by H adsorption, with current increasing again due to hydrogen production near –1.135 V_{MSE} [35,36]. This phenomenon was similarly observed at lower concentrations of K_3IrCl_6, such as 1.0 and 0.5 mM, despite the differences in the current magnitude.

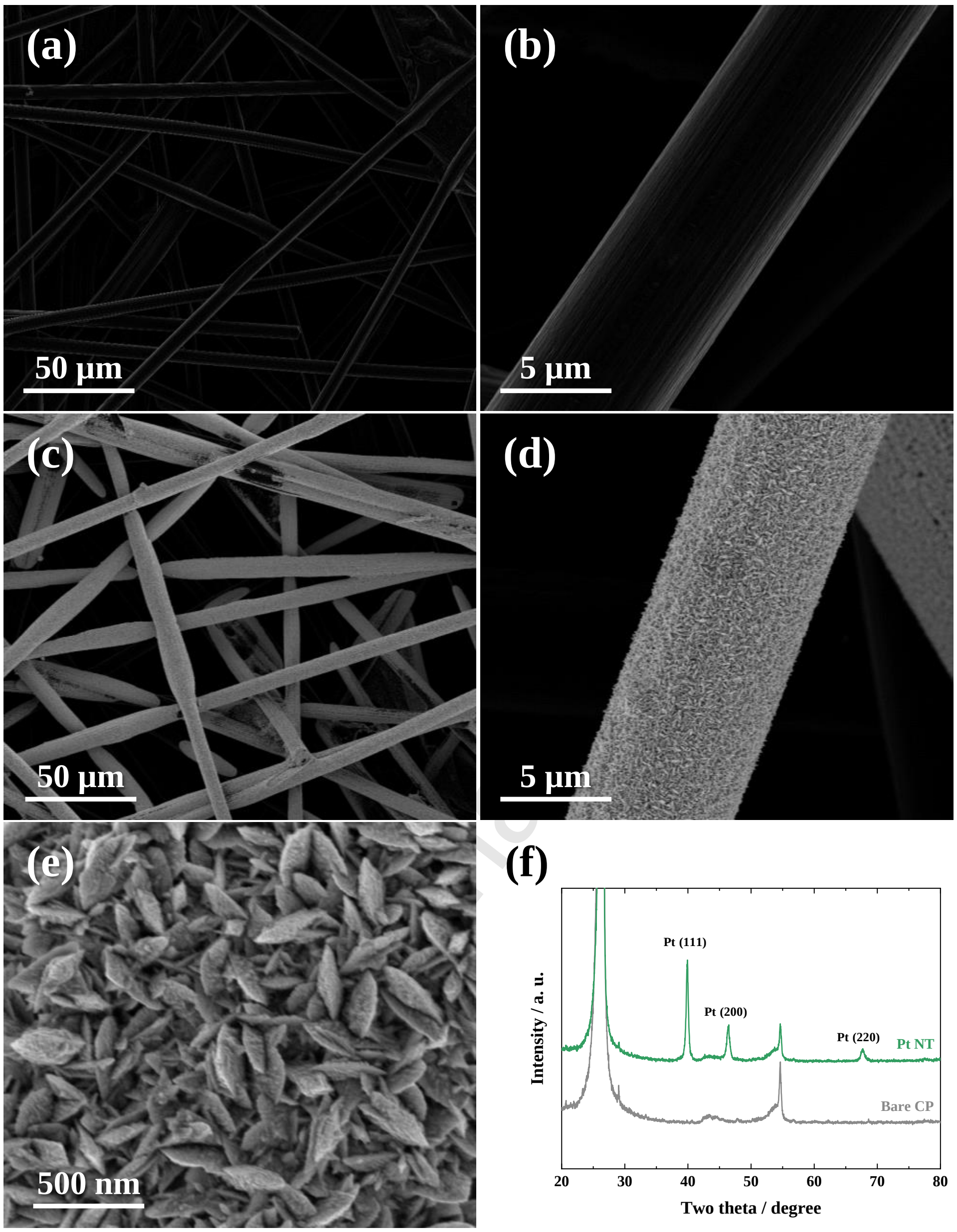

Figure 1. Microscopic visualization and bulk crystallography of the Pt electrodeposit on CP. FESEM images of (a, b) bare CP and (c, d, e) Pt NT acquired at different magnifications. (f) XRD patterns of bare CP and Pt NT.

To further explore Ir SED on the Pt surface, deposition with repetitive potential pulsing was carried out on a Pt NT substrate. Ir was deposited using a single potential pulse while varying the electrodeposition potential (E_{dep}), and the resulting surfaces were characterized by CV in alkaline electrolyte (**Figure S2**).

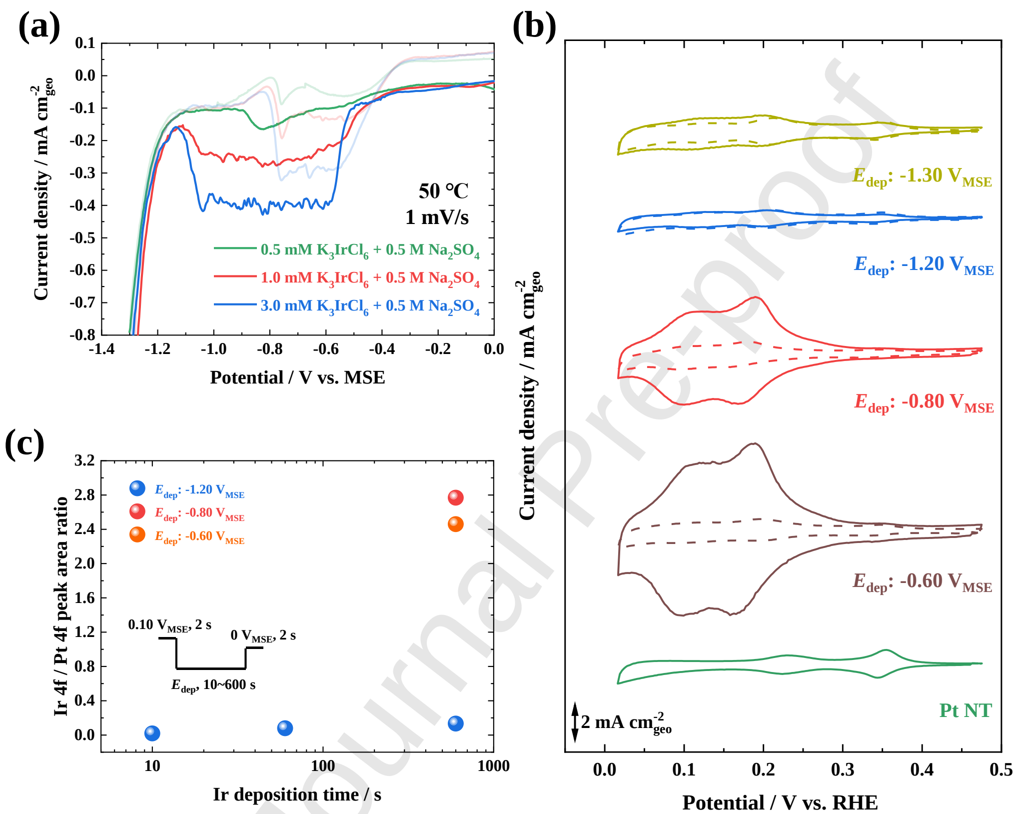

Figure 2. Feasibility of Ir SED on Pt NT. (a) CV curve of Pt NT recorded at 1 mV s^{-1} and 50 °C in Ir deposition electrolyte. (b) H_{UPD} CV profiles of Ir on Pt NT deposited at various E_{dep} in 1.0 mM K_3IrCl_6. The solid lines in the CV profiles represent 600 s, and the dashed lines correspond to 100 s. The CV curves were measured at 50 mV/s in Ar-purged 1.0 M KOH. (c) Ir 4f/Pt 4f peak area ratio of Ir on Pt NT deposited at various E_{dep} in 0.5 mM K_3IrCl_6.

At relatively positive potentials (–0.60 and –0.80 V_{MSE}), a substantial increase in H_{UPD} charge density was observed, exceeding three times that of Pt NT (**Figure S2c)**. In contrast, at more negative E_{dep} values (–1.20 and –1.30 V_{MSE}), the CV profiles showed minimal change (**Figures 2b**). In particular, at –1.20 V_{MSE}, both the H_{UPD} CV profiles and the calculated charge density remained essentially invariant with deposition time.

XPS analysis further confirms the suppression of Ir deposition at negative potentials (**Figure S3**). At –1.20 V_{MSE}, both Pt 4f and Ir 4f signals remained nearly unchanged with increasing deposition time, whereas at less negative potentials, Ir signals increased substantially with a corresponding decrease in Pt signal (**Figure 2c**). The significantly lower Ir 4f/Pt 4f ratio at t –1.20 V_{MSE} indicates that Ir^{3+} reduction is strongly suppressed by H passivation [34–37]. These trends are consistent with the morphology observed by FESEM (**Figure S4**). Ir deposition was also attempted on bare carbon surfaces without a Pt substrate. Even after applying –1.0 V_{MSE} for 600 s, no Ir 4f signal was detected in the XPS analysis (not shown). These results underscore the crucial role of Pt NT in Ir SED, as slow Ir^{3+} reduction appears to be catalyzed by H_{UPD} species on the Pt surface [47].

3.2. Fabrication and characterization of Ir decorated Pt electrode

Building upon the deposition characteristics observed under single-pulse conditions, a repetitive pulsing protocol was implemented to achieve atomic-scale tuning of Ir coverage. As shown in **Figure S4**, each pulse consisted of –1.20 V_{MSE} for 10 s, where Ir^{3+} reduction was suppressed, followed by 0 V_{MSE} for 2 s to reactivate the surface. This sequence, including Ir deposition, H passivation, and surface reactivation, was repeated up to 50 pulses. The resulting samples were denoted as Ir#/Pt NT, where # represents the number of deposition pulses. FESEM images revealed that the overall morphology of Ir-decorated Pt NT remained virtually unchanged compared to the pristine Pt NT, indicating that Ir nucleation proceeded without disrupting the nanosheet architecture (**Figure S5**). To resolve atomic-scale structural modifications induced by Ir decoration, HRTEM was performed on the Pt surface before and after 50 Ir pulses.

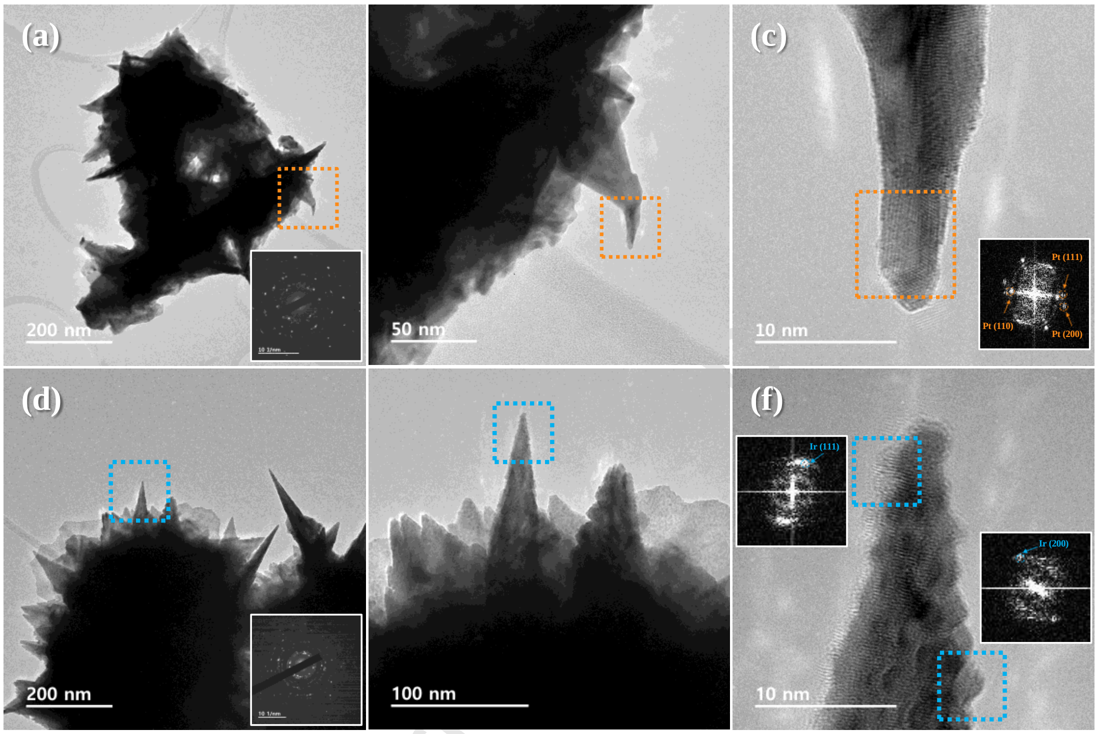

Figure 3. Morphology and crystal structure of Pt NT and Ir50/Pt NT. (a, b) TEM images of Pt NT recorded at different magnifications. Inset: SAED pattern. (c) HRTEM image of Pt NT. Inset: FFT patterns for the marked areas. (d, e) TEM images of Ir50/Pt NT recorded at different magnifications. Inset: SAED pattern. (f) HRTEM image of Ir50/Pt NT. Inset: FFT patterns for the marked areas.

Figure 3a presents the overall morphology of the Pt NT, which consists of 50–200 nm nanosheets assembled into hierarchical assemblies, consistent with the FESEM image. The inset SAED patterns confirmed a polycrystalline structure aligned with the XRD pattern (**Figure 1f**).

Similarly, the TEM images of the Ir50/Pt NT showed virtually identical morphologies and polycrystalline SAED patterns. In contrast, HRTEM images of Ir50/Pt NT (**Figure 3e,f**) revealed that 2–3 nm Ir nanoparticles were distributed on the Pt surface, clearly distinguishing it from Pt NT (**Figure 3b,c**). The presence of Ir was further validated by d-spacing analysis using fast Fourier transform (FFT). For pristine Pt-NT (**Figure 3c**), the d-spacing values matched those obtained from XRD: 0.277 nm for Pt (110), 0.226 nm for Pt (111), and 0.192 nm for Pt (200).

On the Ir50/Pt NT surface (**Figure 3f**), additional d-spacing values corresponding to Ir were observed. These included 0.188 nm for Ir(200) and 0.221 nm for Ir(111), which were not present in the pristine Pt NT. Attempts to confirm the elemental composition using EDS mapping of the TEM images (not shown), did not clearly resolve the Ir signal due to the similar atomic numbers of Pt and Ir.

To probe the electrocatalytic characteristics of Ir#/Pt NT, CV was conducted in 0.5 M H_2SO_4 at a scan rate of 50 mV/s. Initial scans focused on the H_{UPD} region (**Figure 4a**), followed by extended scans encompassing the oxidation potentials of both Pt and Ir (**Figure 4b**).

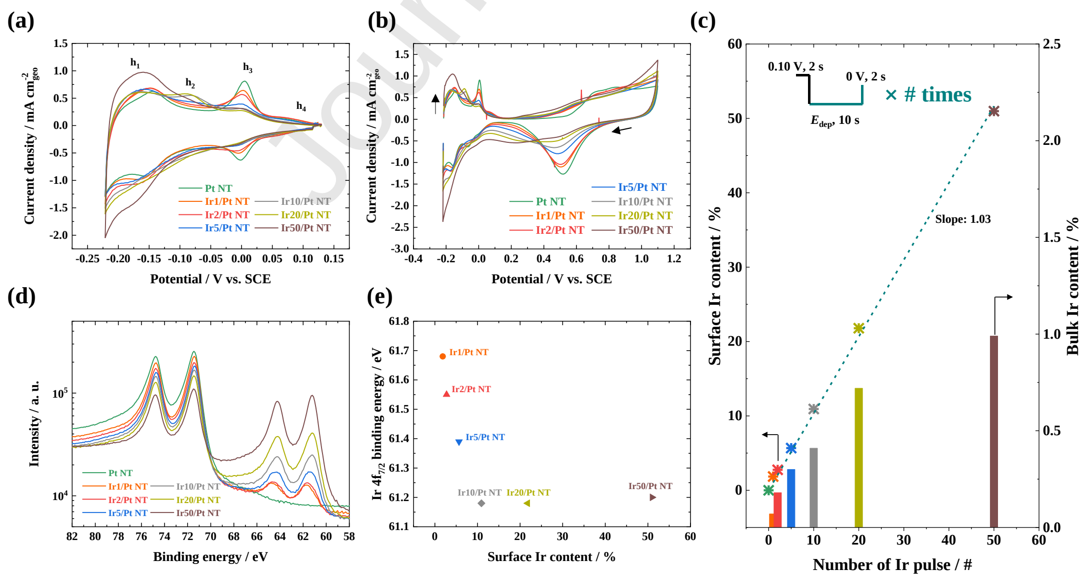

Figure 4. Structural and electronic characterization of Ir-decorated Pt NT electrodes. (a) CV

curves of Pt NT and Ir#/Pt NT recorded at 50 mV/s in Ar-purged 0.5 M H_2SO_4. (b) CV measured immediately after (a) over a wider potential range, starting with an anodic scan. (c) Surface and bulk Ir content of Ir#/Pt NT as a function of the number of Ir pulse. (d) Pt 4f and Ir 4f spectra of Pt NT and Ir#/Pt NT. (e) Ir $4f_{7/2}$ binding energy as a function of surface Ir content.

Figure 4a demonstrates CV curves of Pt NT, with three H desorption peaks observed at approximately –0.14 V_{SCE} (h_1), 0.01 V_{SCE} (h_3), and 0.07 V_{SCE} (h_4). These peaks correspond to the Pt (110), Pt (100), and Pt (100) terraces, respectively, and reflect the H_{UPD} features of highly oriented {100} Pt deposits [32,48,49]. Following 1–2 Ir SED pulses, the Pt (100) step peak (h_3) and Pt (110) peak (h_1) shifted negatively, and the intensity of h_3 decreased significantly. As the number of Ir pulses exceeded 5, the h_4 peak assigned to Pt (100) terrace, which had previously remained stable, disappeared. Simultaneously, a new peak (h_2) at –0.08 V_{SCE} attributed to H_{UPD} desorption of Ir (100), appeared and became increasingly pronounced. After 50 pulses, the CV features resembled that of an Ir (110) single crystal [50]. Further increases in Ir depoosition only led to an increase in the current density without altering the overall CV profile (**Figure S6**).

In the broader potential window (**Figure 4b**), H adsorption features remained consistent with those in **Figure 4a**, although minor differences were observed. For Pt NT, a oxidation and reduction peaks appeared at approximately 0.80 V_{SCE} and 0.5 V_{SCE}, respectively. With increasing Ir coverage, the Pt shifted negatively and diminished, while a new reduction peak near 0.2 V_{SCE} corresponding to oxidized Ir, increased progressively [36]. Electrodes with more than 100 pulses exhibited higher current densities without further changes in profile, indicating complete coverage of the Pt surface by Ir. These electrochemical results indicate that Ir deposition on {100}-oriented Pt NT initiates at Pt (100) step sites, followed by terrace occupation. With sufficient pulsing, a surface structure resembling Ir (110) is formed.

In our previous work on Ir SED on Au substrates, Ir coverage was estimated from the decrease in the AuO_x reduction peak as the number of Ir pulses increased [35,36]. However, as shown in **Figure 4b**, determining Ir coverage on Pt is complicated by the overlap of Pt and Ir redox peaks. Assuming negligible interference from a single Ir pulse, the coverage for Ir1/Pt NT was calculated from the charge integrated between 0.9 and 0.5 V_{SCE}, representing half of the Pt oxide reduction peak, to improve selectivity. This method yielded an Ir coverage of 3.6%, which is less than half of the 8.1% obtained from a single pulse on porous Au dendrites. The surface and bulk compositions of Ir#/Pt-NT were characterized by XPS and ICP-MS, respectively (**Figure 3c**). As shown in **Figure 3c**, the surface Ir content increased proportionally with the number of deposition pulses, starting at 1.85% for a single pulse and reaching 50.97% for Ir50/Pt-NT, with a slope of 1.03. In contrast, bulk Ir content increased monotonically but remained below 1 wt%, even after 50 pulses. For example, Ir5/Pt NT contained 0.96 μg/cm^2 of Ir (0.30 wt%) compared to 319.47 μg/cm^2 of Pt, highlighting the ultra-trace nature of the Ir loading.

The XRD patterns of the Ir#/Pt NT electrodes showed no structural changes compared to those of the Pt NT substrate. No peak shifts were observed even at enlarged scales (**Figure S7**), indicating the absence of bulk alloying between Pt and Ir. In contrast, surface analysis using XPS revealed that the electronic structure of Ir was influenced by the Pt NT substrate, particularly at a low Ir coverage (Ir1–5/Pt NT) (**Figure 4d,e**). For the Ir10–50/Pt NT, the Ir $4f_{7/2}$ peak was positioned at 61.2 eV, which is consistent with metallic Ir^0 [51]. As the Ir content decreased, this peak shifted progressively to higher binding energies, with Ir1/Pt NT exhibiting the Ir $4f_{7/2}$ peak at 61.7 eV, suggesting the presence of partially oxidized Ir species ($Ir^{\delta+}$) [51]. Notably, the Ir5/Pt NT sample exhibits a broadened Ir 4f feature (**Figure 4d**), indicating the coexistence of metallic and oxidized Ir species, consistent with a mixed valence state. In contrast, the Pt $4f_{7/2}$ peak remains at ~71.46 eV across all samples, indicating that Pt remains predominantly in a metallic state without significant change in oxidation state [52]. This shift indicates electronic interaction or surface alloying between Ir and Pt. This shift is attributed to electron depletion from Ir atoms at the Pt interface, likely due to their low coordination and ultralow coverage, which enhanced interfacial effects. This trend is consistent with the higher electronegativity of Pt (2.28) compared to Ir (2.20). Such effects are particularly pronounced at the Ir–Pt boundary, where Ir atoms are directly coordinated with Pt. As the Ir coverage increased, Ir atoms deposited atop existing Ir layers were less affected by the substrate, leading to a binding energy closer to that of bulk Ir. These findings indicate that the electronic structure of Ir is modulated not only by interfacial density with Pt but also by the local electron density distribution within Ir islands.

3.3. Evaluation of AOR performance of Ir#/Pt NT

The electrocatalytic properties of the electrodes were investigated using CV under alkaline conditions. Each sample was measured at least three times to ensure reproducibility. Prior to NH_3 exposure, the H_{UPD} features were recorded in 1.0 M KOH (**Figure S8**), and the resulting CV profiles exhibited trends similar to those observed in acidic media (**Figure 4a**). Upon addition of 0.1 M NH_3, all the electrodes exhibited attenuated H_{UPD} features (**Figure 5a**), consistent with weakened hydrogen–metal interactions induced by NH_3 adsorption [17,53]. A characteristic dehydrogenation peak, associated with the NH_3-to-NH_2 transition at Pt (100) sites was observed at approximately 0.275 V_{RHE} [54,55]. This feature is consistent with previous studies on Pt surfaces, where a peak in a similar potential region has been attributed to the formation of a stable NH_2 adlayer via the initial dehydrogenation of NH_3. Increasing Ir deposition led to a monotonic decrease in AOR onset potential, from ~0.40 V_{RHE} for Pt NT to a minimum of 0.35 V_{RHE} after 20 or more Ir SED pulses. This shift is attributed to the stronger interaction between Ir and *N compared to Pt and *N [24–28]. Additionally, the current density trends for the AOR varied across different potential regions (**Figure 5**). Notably, the reaction mechanism of the AOR on the Pt (100) surface was potential-dependent [54,56].

Two primary pathways have been proposed for the AOR mechanism. On the Pt (100) surface, the N-N pathway becomes predominant at moderate potentials (> 0.5 V_{RHE}) [56]. This pathway involves successive dehydrogenation of $*NH_3$ by OH^- into *N, followed by N-N coupling to yield N_2 [57]. At lower potentials (≤ 0.5 V_{RHE}), the Gerischer–Mauerer (G-M) mechanism dominates [25,56], involving NH_x-NH_y coupling (x = 1 or 2, y = 1 or 2) followed by dehydrogenation to generate N_2 [12,58]. Recent findings suggest that oxidized nitrogen species may act as intermediates in N_2 production at higher potentials (> 0.6 V_{RHE}) [54].

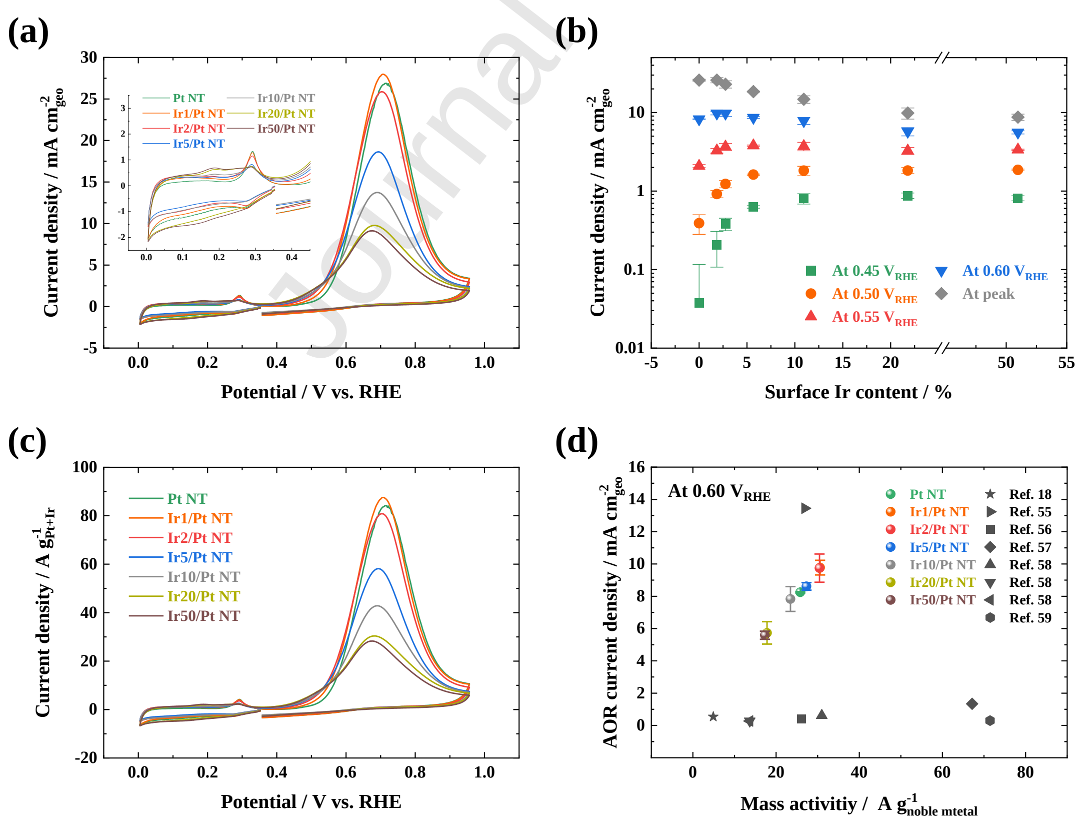

Figure 5. Electrocatalytic evaluation of Ir-decorated Pt nanotube electrodes. (a) CV curves of Ir#/Pt NT electrodes in 0.1 M NH_3 + 1.0 M KOH at a scan rate of 50 mV/s (inset: enlarged H_{UPD} region). (b) Potential-dependent AOR current density of Ir#/Pt NT as a function of surface Ir content. (c) Geometric AOR current density versus surface Ir content. (d) Benchmarking of mass and geometric AOR activities of Ir#/Pt NT electrodes against reported Pt-based catalysts.

To quantify these potential-dependent effects, the AOR current density was extracted from the CV profiles in **Figure 5a** and plotted against the surface Ir content (**Figure 5b**). In the lower potential range (≤ 0.5 V_{RHE}), where the G-M mechanism is dominant [25,56], the incorporation of Ir into Pt (100) framework significantly alters this mechanistic landscape. At low Ir surface content (≤ 5.67%), the current density in the low-potential regime increases sharply, consistent with the promotion of NH_x-NH_y coupling via isolated Ir sites. However, as the Ir coverage increases beyond this threshold, the current density begins to decline Specifically, in higher potential regions (> 0.6 V_{RHE}), even a small increase in Ir content above 5% led to a notable reduction in the current density. This behavior is attributed to strong *N adsorption on the electrode surface, which suppresses N-N coupling by blocking the formation of N_{ads}, effectively poisoning the active sites [25]. The formation of NO_x intermediates under these conditions further exacerbates performance degradation. Recent studies have shown that limiting the upper potential to below 0.75 V_{cell} during AOR evaluation suppresses electrode oxidation and is crucial for long-term catalytic stability [59]. Under these practically relevant conditions (≤ 0.60 V_{RHE}), all Ir#/Pt NT electrodes with up to 5.67% surface Ir content, consistently outperformed pristine Pt, demonstrating enhanced AOR activity within the stable operating range. These trends were consistently observed across different alkaline conditions, as confirmed by measurements at varying KOH concentrations (**Figure S11**).

The trends observed in mass activity measurements were broadly consistent with the geometric AOR performance. Because the Ir loading remained extremely low relative to the Pt mass, the mass activities of Ir#/Pt-NT increased with pulse number and paralleled the geometric current density profiles (**Figure 5c**). At 0.60 V_{RHE}, the Ir#/Pt NT electrodes exhibited mass activities comparable to those of state-of-the-art Pt-based AOR catalysts (**Figure 5d**) [18,59–63]. The reported activities were significantly influenced by testing conditions, including scan rate, rotation speed, electrolyte pH, and NH_3 concentration [64,65]. A detailed comparison of these parameters and catalyst structures is summarized in **Table S1**. While some powder-based catalysts have reported higher mass activities, often as a result of lower catalyst loadings, their geometric activities were generally modest, which may limit their practical relevance. In comparison, the Ir#/Pt NT electrodes, prepared with controlled Ir deposition, exhibited strong geometric performance under similar testing conditions.

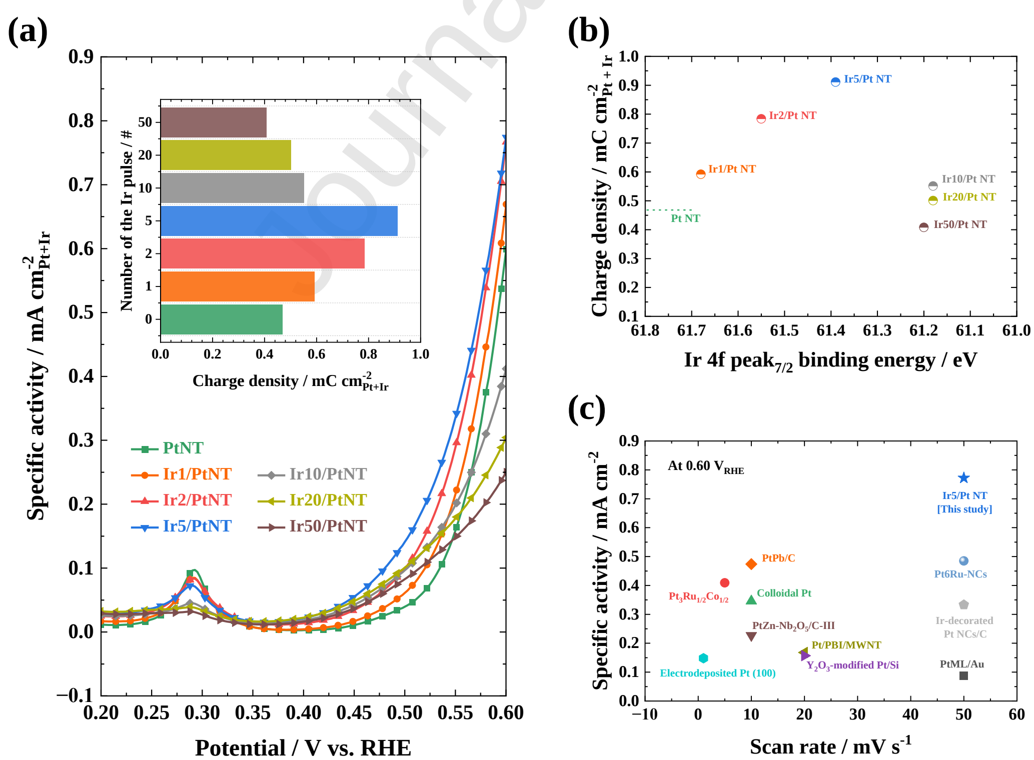

Figure 6. Specific AOR activity of Ir-decorated Pt NT electrodes. (a) CV-derived specific activities of Ir#/Pt NT electrodes. (b) Integrated AOR charge density of Ir#/Pt NT electrodes as a function of Ir $4f_{7/2}$ binding energy. (c) Benchmarking of Ir5/Pt NT against reported Pt-based AOR catalysts at 0.60 V_{RHE} under different scan rates.

To evaluate the intrinsic AOR behavior under practically relevant conditions up to 0.60 V_{RHE}, we normalized the current densities by the surface roughness factor. **Figure 6a** shows that all Ir-decorated Pt NT electrodes exhibited higher specific activity than pristine Pt NT below 0.55 V_{RHE}. The trend mirrors the trade-off observed in geometric activity, but current density alone does not fully capture the potential dependence of AOR performance. To complement this, the integrated AOR charge from the full CV range was also analyzed (**inset of Figure 6a**). This provides a broader view of the cumulative catalytic response across the kinetically relevant potential window. The integrated AOR charge exhibited a volcano-type dependence on Ir coverage and reached a maximum at Ir5/Pt NT, which delivered nearly double the charge of the pristine Pt NT.

As discussed earlier, Ir deposition up to five pulses mainly targets the Pt (100) step sites, as confirmed by H_{UPD} and TEM analysis. XPS data showed a progressive shift in the Ir $4f_{7/2}$ binding energy up to Ir5/Pt NT, beyond which the shift saturates around 61.2 eV, consistent with the onset of inter-Ir coordination. This transition is mirrored in **Figure 6b**, where both the binding energy and integrated charge rise coherently up to Ir5/Pt NT, then diverge sharply. The peak in specific activity emerges right before this electronic saturation point, reinforcing the idea that Ir remains electronically coupled to Pt just before Ir atoms start forming clusters that weaken Pt interaction. **Figure 6c** benchmarks the specific activity of Ir5/Pt NT at 0.60 V_{RHE} against state-of-the-art Pt-based AOR catalysts as a function of scan rate (**Table S2**) [29,33,66–73]. Although the Ir5/Pt NT performance was evaluated at a relatively high scan rate, it remains among the best-in-class (50 mV/s), underscoring the catalytic advantages of finely tuned interfacial Ir–Pt configurations.

3.4. Theoretical investigation of AOR mechanism on Ir modified Pt surfaces

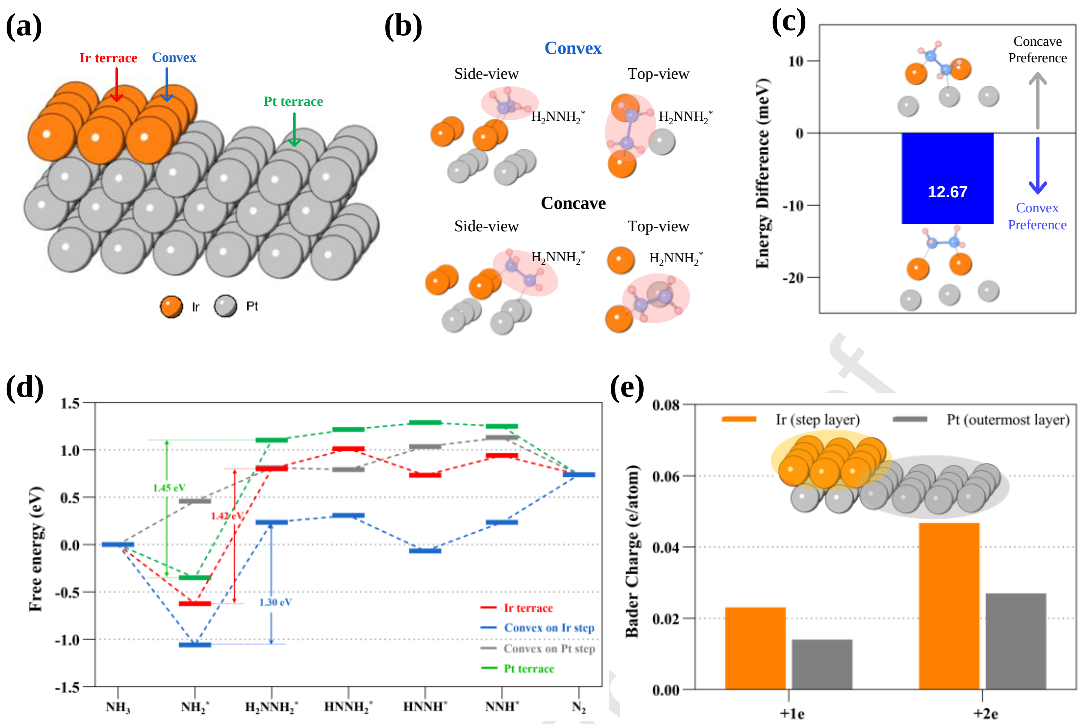

Figure 7. DFT analysis of adsorption geometry, reaction energetics, and charge distribution at Ir–Pt step interfaces. (a) Schematic illustration of three representative adsorption sites for AOR on an Ir step supported on a Pt (100) surface: Ir terrace, Ir–Pt interface, and Pt terrace. (b) Adsorption structures of H_2NNH_2* on convex and concave Ir–Pt sites, and (c) the corresponding energy difference between the two configurations. (d) Gibbs free energy diagrams of the AOR pathways at different adsorption sites on slab model. (e) Bader charge per atom on the Ir step layer (12 atoms) and outmost Pt layer (16 atoms) upon addition of extra electrons.

DFT calculations were performed to elucidate the origin of the AOR observed at low Ir coverages on Pt (100). Since the AOR enhancement by Ir was most pronounced at low-overpotential (≤ 0.5 V_{RHE}), where the G-M mechanism dominates, this pathway was chosen for further theoretical investigation [74]. As illustrated in **Figure 7a**, three representative adsorption sites on step surface model were evaluated: (i) a pristine Pt terrace; (ii) an Ir terrace deposited on the Pt(100) surface and (iii) an Ir-Pt interface formed by Ir atoms at step edges. To probe the preferential adsorption site between convex and concave geometries, the adsorption behavior of H_2NNH_2 was evaluated on each site. **Figure 7b** presents the optimized adsorption geometries of the H_2NNH_2 intermediates formed via NH_2-NH_2 coupling, which has been reported as the rate-determining step of the G-M mechanism on Pt (100) and Ir (100) surfaces [75]. Although H_2NNH_2 can adsorb on both concave and convex sites, convex site was thermodynamically more favorable and selected for further analysis (**Figure 7c**).

The calculated Gibbs free energy diagrams of G-M mechanism at each site are presented in **Figure 7d**. Among the three adsorption sites, the convex Ir-Pt interface exhibits the strongest thermodynamic driving force for the initial dehydrogenation step, favoring the formation of NH_2^* from NH_3.

The free energy change for coupling of two NH_2 to H_2NNH_2, identified as the rate-determining step across all three sites, decreases from 1.45 eV on Pt to 1.42 eV at the Ir terrace and further to 1.30 eV at the Ir–Pt interface. While both Ir-containing sites lower the reaction barrier, the 0.12 eV difference highlights the diminishing catalytic benefits of excessive Ir coverage. This decrease in energy barrier reflects the beneficial role of Ir incorporation and the step-sites formation facilitating the rate-determining AOR step. These computational results align with experimental results, which collectively indicate that AOR activity peaks when Ir atoms are confined to step sites and maintain strong electronic coupling with the Pt substrate, as exemplified by the Ir5/Pt-NT sample.

To further validate the superior catalytic performance of the Ir step site, a Pt step model was evaluated as a reference. The atomic configurations of AOR intermediates at the Pt step site are shown in **Figure S10**. NH_2^* formation from NH_3 on the Pt step was found to be endothermic, indicating that the step site is not catalytically active. In contrast, the Pt terrace is likely to serve as the preferred site for AOR for the pure Pt catalyst. In addition, to evaluate the initial adsorption energies for ammonia at each site, we calculated the NH_3 adsorption energies assuming that molecular NH_3 first adsorbs on the surface prior to the dehydrogenation step (**Figure S13**). The results show a clear site-dependent trend: Ir step (−2.02 eV) > Ir terrace (−1.71 eV) > Pt step (−1.69 eV) > Pt terrace (−1.61 eV), supporting that Ir decoration provides enhanced adsorption capability for ammonia. These results collectively support that the presence of Ir decoration at Pt (100) sites facilitates the catalytic activity of AOR by stabilizing NH_2 adsorption and lowering the energy barrier for the rate-determining step.

To probe the electron affinity of Ir-decorated step on Pt surface during the AOR, Bader charge analysis was performed with extra electrons into the slab models (**Figure 7e**). The Ir step site consistently accommodates more than twice those observed at neighboring Pt atoms reflecting the stronger electron affinity of the Ir site. In each step of the AOR, each adsorbed N_xH_y intermediate is deprotonated by OH^- to form the next species and H_2O, releasing one electron into the electrode. The preferential accumulation of these electrons at Ir step sites suggests that Ir atoms can serve as efficient electron sinks, facilitating post-reaction electron uptake.

3.5. MEA-Based Ammonia Electrolysis Performance

To evaluate the practical applicability of the Ir-decorated Pt NT electrodes, their ammonia electrolysis performance was further investigated in a MEA-type ammonia electrolyzer. Ir5/Pt NT and Pt NT were employed as the anode catalysts, whereas a 40 wt% Pt/C catalyst supported on carbon paper was used as the cathode. Electrolyzer performance was investigated under both potentiodynamic and galvanostatic conditions at 80 °C using 1.0 M KOH (catholyte) and 1.0 M KOH + 1.0 M NH_4OH (anolyte).

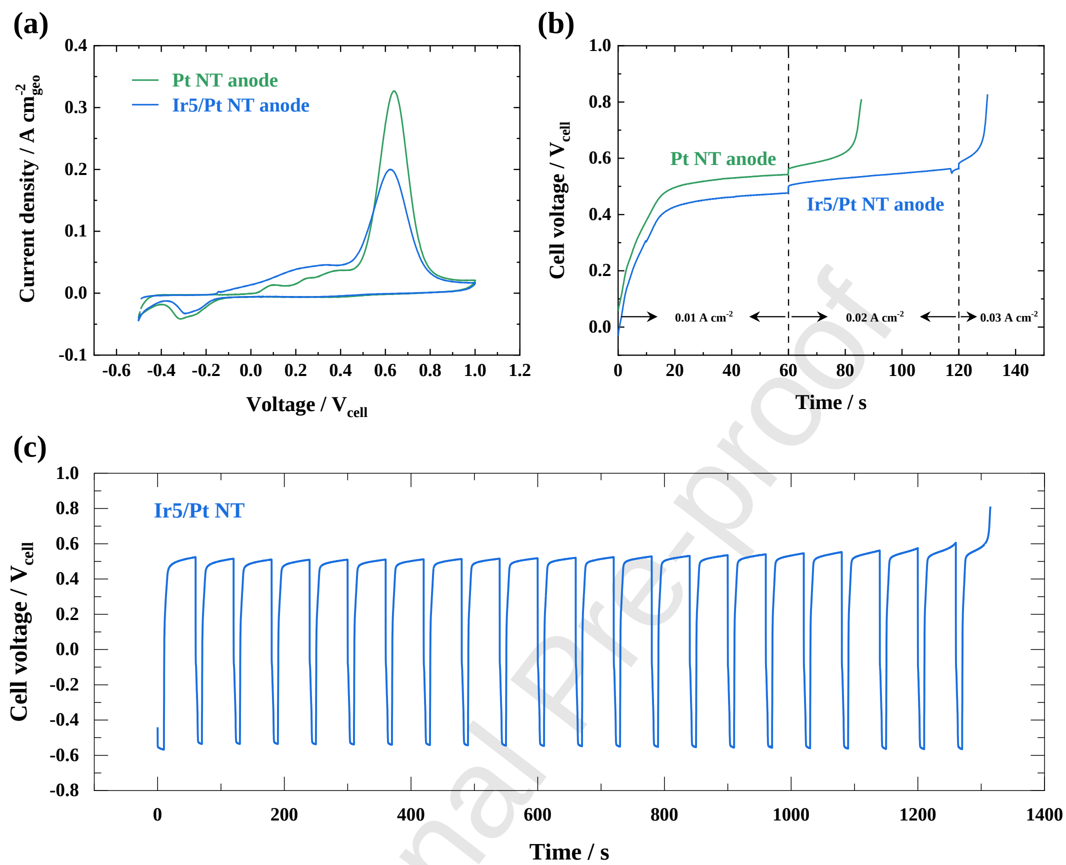

Figure 8. Performance of Ir5/Pt NT and Pt NT anodes in an MEA-type ammonia electrolyzer at 80 °C. (a) Potentiodynamic polarization curves. (b) Galvanostatic step measurements at 10, 20, and 30 mA cm^{-2}. (c) Pulsed electrolysis of Ir5/Pt NT with alternating steps consisting of an AOR step at 20 mA cm^{-2} for 50 s and reactivation step at −0.05 mA cm^{-2} for 10 s.

The MEA-based electrolyzer results exhibit a clear potential-dependent behavior consistent with the half-cell measurements (**Figure 8a**). At lower cell voltages, the Ir5/Pt NT anode delivers higher current densities than Pt NT, whereas at higher voltages, Pt NT shows comparatively higher current response. The enhanced performance of Ir5/Pt NT in the low-voltage region can be attributed to the stronger NH_3 adsorption, as well as facilitated NH_2–NH_2 coupling via the G–M mechanism. In contrast, the reduced performance at higher voltages is likely associated with overly strong adsorption of nitrogen-containing intermediates, leading to accelerated surface poisoning. In addition, contributions from Pt oxidation at higher potentials may partially account for the increased current observed on Pt NT, as its poor stability under galvanostatic operation suggests that the high current response does not solely originate from sustained AOR activity (**Figure 8b**).

Under stepwise current operation (10, 20, and 30 mA cm^{-2}, each for 60 s), Pt NT failed to sustain even 20 mA cm^{-2}, showing a rapid increase in cell voltage. In contrast, Ir5/Pt NT maintained stable operation for a longer duration, although it was still unable to sustain 30 mA cm^{-2}.

To address the instability arising from surface poisoning, pulsed electrolysis was employed by introducing a short reactivation step (−0.05 mA cm^{-2} for 10 s) to reduce accumulated nitrogen species on the electrode surface [13,54,59]. This pulsed operation significantly improved operational stability, allowing the cell to operate for approximately 1300 s before a voltage increase was observed (**Figure 8c**).

To gain insight into surface evolution under electrochemical conditions, post-reaction XPS analysis was performed. Direct analysis of electrodes from the MEA-type electrolyzer was not feasible due to catalyst layer adhesion to the membrane during disassembly, so controlled half-cell measurements were employed (**Figure S14**). After reaction, both Pt 4f and Ir 4f peaks shift toward higher binding energy, accompanied by an increase in the O 1s signal, indicating surface oxidation. Notably, such changes are observed even under relatively mild electrochemical conditions. In addition, the Ir $4f_{7/2}$ peak becomes narrower after reaction, suggesting a transition toward a more uniform oxidized surface state that is not fully reduced even after a reductive step. These results indicate that the catalyst surface undergoes dynamic chemical changes under operating conditions.

While pulsed electrolysis extends the operational duration compared to constant current operation, further optimization of cell configuration, pulsing protocols, and electrode architecture is still required to achieve sustained long-term performance and fully establish catalytic applicability. These results provide a proof-of-concept demonstration of the catalyst under device-relevant conditions.

4. Conclusions

This study presents a self-terminated electrodeposition strategy for atomic-scale Ir decoration of facet-controlled Pt (100) nanosheet electrodes with ultra-low loadings. Ir deposition initiates at step sites under hydrogen-passivated conditions, enabling precise control of Ir coverage from 1.85% to 50.97% while maintaining bulk loadings below 1 wt%.

Electrochemical evaluation under alkaline ammonia oxidation revealed that Ir decoration progressively lowered the onset potential relative to pristine Pt. At low coverages, this effect was accompanied by enhanced activity, while excessive Ir led to poisoning and reduced current density at higher potentials. As a result, both potential-dependent current density and integrated charge followed a volcano-type dependence, with Ir5/Pt NT (5.67% surface Ir) achieving the highest specific activity of 0.772 mA/cm^2 at 0.60 V_{RHE}, nearly doubling that of pristine Pt NT. Consistently, the integrated AOR charge density was nearly twice that of pristine Pt NT, highlighting the cumulative enhancement over the full potential window. Importantly, this optimal performance was realized with only 0.96 $\mu g/cm^2$ of Ir, demonstrating the efficiency of atomic-scale utilization.

The superior performance of Ir5/Pt NT is attributed to step-selective Ir decoration. H_{UPD} confirmed preferential occupation of step sites, while XPS revealed binding energy shifts consistent with strong Pt–Ir coupling. Beyond five pulses, Ir deposition extended onto terraces and formed Ir–Ir domains, disrupting this favorable configuration and explaining the decline in activity. DFT calculations further identified the $*NH_2$-$*NH_2$ coupling step as the rate-determining step of the AOR, with its free energy barrier decreasing from 1.45 eV on Pt to 1.30 eV at the Ir–Pt step interface. Bader charge analysis further supports enhanced electron affinity at the Ir step, which provides more favorable electronic environment for ammonia oxidation.

In addition, evaluation in a MEA-type ammonia electrolyzer confirms improved performance of the Ir5/Pt NT anode at low cell voltages, supporting its practical applicability.

Collectively, these results demonstrate that isolated Ir decoration at Pt step sites optimally enhances AOR activity, while higher Ir coverage undermines this effect. This work links atomic-scale interfacial structure to catalytic performance and provides guidance for the rational design of heteroatomic catalysts in ammonia energy conversion.

CRediT authorship contribution statement

Hyunki Kim: Methodology, Investigation, Data curation, Conceptualization, and Writing – original draft. Daehyun Kim: Methodology, Investigation, and Writing – original draft. Junbeom Bang: Investigation, Data curation, Seokjin Hong: Investigation, Data curation. Haesun Park: Methodology, Supervision, Writing – review and editing, Project administration. Sang Hyun Ahn: Methodology, Supervision, Writing – review and editing, Project administration.

Declaration of competing interests

The authors declare that they have no competing financial interests or personal relationships that may have influenced the work reported in this paper.

Acknowledgments

This work was supported by National Research Foundation of Korea (NRF) grants funded by the Korean government (MSIT) (Nos. RS-2025-02304646, RS-2022-NR068485, RS-2024-00413272, 2022R1A6A3A13063678). Hyunki Kim, Daehyun Kim, and Junbeom Bang contributed equally to this work.

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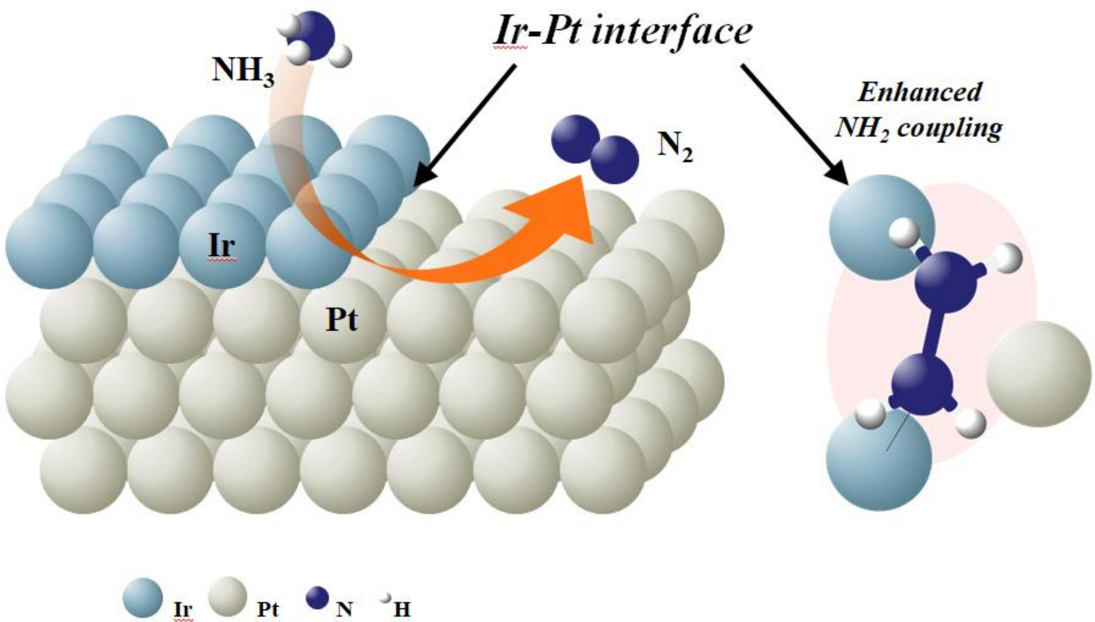
Graphical abstract



Highlights

- Self-terminated electrodeposition enables atomic-scale Ir decoration on Pt(100) with ultra-low loading.
- Ir deposition preferentially targets Pt(100) step sites, forming confined Ir domains that remain electronically coupled to Pt.
- Step-confined Ir–Pt interfaces enhance ammonia oxidation and nearly double the catalytic charge relative to pristine Pt.
- Higher Ir loading produces Ir–Ir domains that disrupt interfacial coupling and reduce catalytic activity.
- DFT reveals that Ir–Pt step sites lower the NH_2 coupling barrier and facilitate electron redistribution during AOR.